

Jumps, Impulses, and Convolutions

This section covers some notation that is common in computational neuroscience for treating spikes as abrupt jumps.. As of 2025, most of this content is not *directly* assessed, with an important exception: Students will be expected to be able to describe dynamical systems that jump when a spike arrives using mathematics. This will probably use some of the notation described in this section.

Hybrid systems with jumps

Imagine modelling the concentration $x = [A]$ of a neurotransmitter in the synaptic cleft, which decays at a constant rate $1/\tau$, and increases by amount q every time the presynaptic cell spikes at time t_s . If we provide the set of spike times $t_s \in \mathcal{T}$, then we can describe this as an ODE with an explicit jump condition:

$$\begin{aligned} \tau \dot{x}(t) &= -x(t) & \text{if } t \notin \mathcal{T} \\ x(t^+) &\leftarrow x(t^-) + q & \text{if } t \in \mathcal{T} \end{aligned} \quad (1)$$

The notation $x(t^-)$ and $x(t^+)$ splits the points where $x(t)$ is discontinuous into a before/after value.

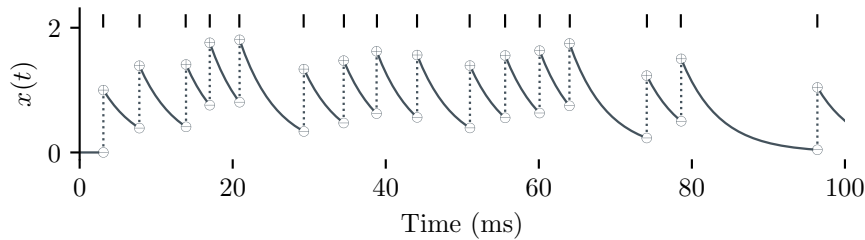


Figure 1: Solution to a model for neurotransmitter concentration in the synaptic cleft: This plot shows a solution to the time-varying concentration $x(t)$ from (1) using $\tau = 5$ ms and $q = 1$ (normalized concentration units), with presynaptic spike times indicated by the tic marks along the top of the plot. The annotations $\ominus$ and $\oplus$ mark the left $x(t^-)$ and right $x(t^+)$ values of the solution at discontinuities, shown here for didactic purposes.

Dirac distribution

You may see Equation (1) written as

$$\tau \dot{x} = -x + q \cdot \delta(t - t_s), \quad t_s \in \mathcal{T} \quad (2)$$

where $q \cdot \delta(t - t_s)$ means “add q to x whenever $t = t_s \in \mathcal{T}$ ”. The notation $\delta(t - t_s)$ is a *Dirac delta distribution*. It is infinitely tall, infinitesimally wide, and defined so that its integral is always 1. There are many ways to construct this as a limit. One approach is to use a normal distribution with mean $\mu = t_s$, and consider the limit as its standard deviation goes to zero (Figure 2):

$$\delta(t - t_s) = \lim_{\sigma \downarrow 0} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(t-t_s)^2/\sigma^2}. \quad (3)$$

Computational neuroscience uses the Dirac distribution as shorthand for a quick jump in signals caused by a spikes. This is partly inherited from signal processing and mathematical disciplines that use the nice mathematical properties of $\delta(t - t_s)$ to solve linear systems. Whether this is a good idea is up for debate: You can’t always manipulate $\delta(t - t_s)$ like an ordinary function, and its best to read it as shorthand for a jump as in (1).

Impulse $\delta(t)$ as $\sigma \rightarrow 0$ limit of Gaussian

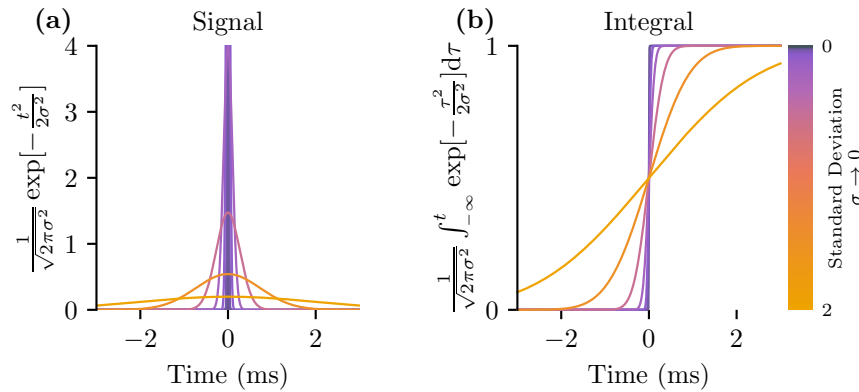


Figure 2: Constructing a Dirac distribution as the zero-variance limit of a Gaussian distribution: **(a):** The normal distribution becomes progressively more concentrated as its standard deviation approaches zero. **(b):** (cumulative distribution function) the distribution’s height diverges in the limit, but its integral is always 1.

Convolutions

If we ignore the spikes and start at $x(0) = 1$, then the linear ODE component of (1) has solution $x(t) = \exp(-t/\tau)$. This is the systems *impulse response*.

We can write solutions to (1) as a sum over impulse responses to each presynaptic t_s , scaled by the quantity q of neurotransmitter released:

$$x(t) = \sum_{t_s \in \mathcal{T}} h(t - t_s), \quad h(t) = q \cdot \exp(-t/\tau). \quad (4)$$

We can also write (4) as a *convolution* of the filter $h(t)$ with an *impulse train* $y(t)$, denoted here with the operator $*$:

$$\begin{aligned} x(t) &= [h * y](t) \\ y(t) &= \sum_{t_s \in \mathcal{T}} \delta(t - t_s), \end{aligned} \quad (5)$$

where, again, $\mathcal{T}$ is a set containing the presynaptic cell's spike time t_s . This is equivalent to (4). More generally, the convolution is defined as an integral:

$$[h * y](t) = \int_0^\infty h(t') y(t - t') dt'. \quad (6)$$

In words, (6) says: To find the value $x(t)$, look into the past with time-lag $t' \in [0, \infty)$; each unit impulse in y at the past time $t - t'$ affects the current time-point t by amount $h(t')$ given by the history filter.

The convolution integral (6) will work to find the linear response to any input $y(t)$, not just impulse trains. You can check that (6) reduces to (5) and (??) by considering a single spike $y(t) = \delta(t - t_s)$, in which case the convolution integral $\int_0^\infty h(t') \delta(t - t' - t_s) dt' = h(t - t_s)$ reduces to evaluating $h(t)$ at time lag $t - t_s$.

Processing time-series data in computational neuroscience uses convolutions extensively. You can find built-in implementations like `np.convolve` in Python. Some implementations use the computational savings of the *fast Fourier transform* and *convolution theorem* to quickly calculate the convolutions of long time-series.